Danesh Morales Hashemi

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Research Positions

2025 – Present **Graduate Research Assistant**, Institute for Quantum Computing
Advisor: Graeme Smith

2024 – 2025 **Undergraduate Research Assistant**, Institute for Quantum Computing
Advisor: Joseph Emerson
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*

Fall 2023 **Undergraduate Research Assistant**, University of Waterloo
Advisor: Roberto Guglielmi
Topic: *Controllability of Fokker-Planck Equation*

Education

2025 – Present **MMath in Applied Mathematics**, University of Waterloo
Quantum Information Specialization

2020 – 2025 **BMath in Mathematical Physics**, University of Waterloo
Co-op Program · Pure Mathematics Minor · Graduated with Dean's Honours

Conferences and Talks

March 2025 **"Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation"**
APS March Meeting 2025, Session on Scalable Verification and Validation (QCVV), Anaheim Convention Center

Teaching Experience

Spring 2025 **Teaching Assistant**, AMATH 250 – Introduction to Differential Equations

Fall 2023 **Instructional Support Assistant**, MATH 115 – Linear Algebra for Engineering

Winter 2022 **Instructional Support Assistant**, CS 115 – Introduction to Computer Science

Projects

December 2025 **"Introduction to Cycle Benchmarking for Quantum Error Diagnostics"**
QIC 820 Final Project, Institute for Quantum Computing

December 2025 **"Predicting MLB Playoffs Using Quantum Machine Learning"**
QIC 890 Final Project, Institute for Quantum Computing. Link

April 2025	**"Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation"** Undergraduate Thesis, University of Waterloo
August 2024	**"Introduction to Superoperator Representations and Characterization of Completely Positive Maps"** AMATH 900 Final Project, University of Waterloo

Awards and Scholarships

Institute for Quantum Computing Undergraduate Research Award 2024	**16,000 CAD**
University of Waterloo President's Scholarship 2020	**2,500 CAD**
Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama	**250,000 USD**

Relevant Coursework (with links to syllabus)

Quantum Physics	Quantum Theory 1/2/3	Representation Theory
Quantum Information Processing	Quantum Machine Learning	Groups and Ring Theory
Quantum Devices	Theory of Quantum Information	Galois and Field Theory